

Explain-and-Test: An Interactive Machine Learning Framework for Exploring Text Embeddings

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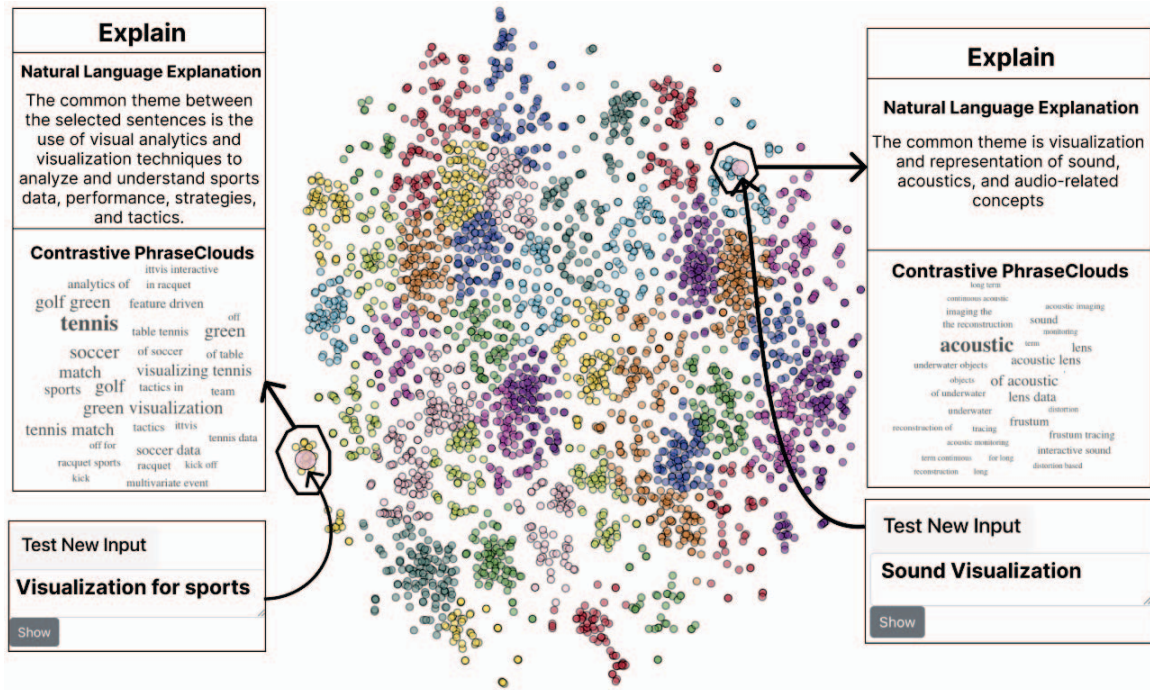


Figure 1: TextCluster Explainer visualizes text embeddings, allowing users to lasso clusters of interest to get automated **explanations** of these clusters. The system provides two kinds of explanations: *Contrastive PhraseClouds* generated by an SVM model and *Natural Language Explanations* generated by a large language model (GPT-4). Users can **test** these automated explanations by entering manual labels to be dynamically embedded into the visualization, a feature we call *Assessment by Re-projection*. If the new text is mapped to the cluster, it validates the automated explanation. The figure shows a t-SNE projection where each point corresponds to the paper title from the Visualization Publications dataset along [13] along with explanations and test inputs for two different clusters. The color encoding corresponds to labels predicted by clustering the projections.

ABSTRACT

Text embeddings—mappings of collections of text to points in high-dimensional space—are a common object of analysis. A classic method to visualize these embeddings is to create a nonlinear projection to two dimensions and look for clusters and other structures in the resulting map. Explaining why certain texts cluster together, however, can be difficult. In this paper, we introduce a human-in-the-loop framework for applying machine learning (ML) to this challenge. The framework has two stages: (1) **explain**, in which we use ML to produce a description of a pattern; and (2) **test**, in which the user can verify the explanation by entering new text that

fits the pattern, and sees where it appears on the map. If the new text is mapped to the original cluster, that is evidence in favor of the ML-generated explanation. We illustrate this process with a visualization application that provides two kinds of explanations: Natural Language Explanations and Contrastive PhraseClouds. Scenarios on exploring academic papers and literary work showcase the benefit of our workflow in discovering related topics and analyzing thematic differences in text.

Index Terms: Text Visualization, Dimensionality Reduction, Clustering—Large Language Models—Explanation—;

1 INTRODUCTION

Text analysis is a widespread challenge. Social media moderators reading online conversations, literary critics practicing distant reading, and lawyers looking through thousands of emails for evidence all need to find structure and meaning in large collections of text. A classic approach to this challenge is to embed documents in a high-dimensional space and use nonlinear projection techniques to create a “map” of the data [4, 14]. Often the resulting map has an

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intricate visual structure, which holds the promise of guiding further exploration.

Unfortunately, making good on this promise is hard. In practice, relating visual structure to meaningful aspects of the underlying data can involve drilling down on multiple data points, attempting to spot patterns, and trial and error. While classic techniques exist for describing the high-level structure of textual datasets, such as topic modeling or keyword extraction, they do not allow for interaction. A crucial idea, however, was introduced in the Clustrophile application [2] for non-textual data. The system let users select sets of points on a visualization and then uses an interpretable machine learning model to explain what those points have in common.

Inspired by this idea, we propose a machine-learning-based system aimed at text exploration. Like Clustrophile, our application lets users select arbitrary subsets of dots in visualization and displays the common patterns they share. We then add two new features. First, we provide an option to query a state-of-the-art AI system (GPT-4) for a natural-language explanation. We find explanations produced by this system to be clear, detailed, and plausible. Of course, language-model-based AI can famously produce “hallucinations” [12, 15]: reasonable-seeming results that turn out to be confabulated. To address this issue, we add a second feature: the ability to type in a new label or piece of text and see where appears in the visualization—this lets the user interactively test whether a manually created label/explanation of a cluster ends up being positioned inside the cluster in question, either validating or disproving the user hypothesis.

These new features enable a rapid two-stage workflow, which we call **explain-and-test**. In the first stage, a user can quickly get an explanation of a cluster in a nonlinear projection. In the second stage, they examine and assess this explanation by adding new, hypothetical data to the visualization. Our goal in this short paper is to describe a working prototype of the explain-and-test framework and to illustrate how it can be used with several scenarios based on real-world data.

2 RELATED WORK

Exploring text by mapping it into a high dimensional vector space is a popular technique [14, 16, 25]. While simple linear models or explainers [7] can be used to craft user-informed projections, we use standard non-linear dimensionality reduction methods to project and visualize the high-dimensional space. This method often reveals an intriguing set of clusters. Understanding what these clusters mean, however, can be surprisingly difficult.

Tools for analyzing high-dimensional projections of other data types are not always applicable to this text-specific clustering problem. Latent Space Cartography [18], for example, focuses on globally meaningful directions in high-dimensional space, rather than local clustering. Tools such as t-viSNE [3] do allow inspection of clusters, but give explanations in terms of coordinates in high-dimensional space, which are not intrinsically meaningful for text embeddings. However, a helpful interaction idea introduced in Clustrophile [2, 5] (and also used in t-viSNE) lets users “lasso” clusters to see an explanation. We have borrowed this idea in our system.

Applications that are tuned for analyzing text embeddings often use machine learning to generate explanations for clusters. For example, DeepClustering [10] fits a logistic regression model to extract keywords associated with a cluster. Correlating pre-existing human labels to clusters [17] can also be helpful. As machine learning technology has advanced, we have seen more sophisticated techniques for generating text embeddings that benefit from natural language understanding capabilities of language models, e.g., BERTopic [9] and KeyBERT [8], which use SentenceBERT [23] to encode the text, but use traditional keyword extraction methods or comparing cosine similarity between the embeddings of text and extracted words to generate keywords.

There is an important tension in ML-generated explanations. On one hand, modern natural language processing seems to be extremely capable; on the other hand, it can be a “black box” process, producing plausible but not necessarily trustworthy results. The prototype system we present addresses this tradeoff: using a state-of-the-art system to provide explanations that potentially go far beyond a list of keywords, while using interactivity to help users calibrate their trust in the technology.

3 TEXTCLUSTER EXPLAINER AND EXPLAIN-AND-TEST FRAMEWORK

Our goal is to explore the capacity of machine learning to provide cluster explanations, and how best to allow users to take advantage of that capacity. Our platform for this exploration is a custom-built prototype, the **TextCluster Explainer**, which lets users analyze embeddings of short snippets of text.

The basic interaction technique of the TextCluster Explainer is simple. The user is presented with a visualization of text embedding. In addition to standard interactions, such as mouseovers for detail, they may use a freeform “lasso tool” to outline any region or cluster of interest. The user can tweak the attributes of the visual encoding like dot size and opacity to identify high-density clustered regions. The TextCluster Explainer will explain what is distinctive about the cluster, using two machine learning methods: a conventional linear classifier, and a state-of-the-art system, GPT-4 [21].

If desired, the user can test the explanation by specifying arbitrary text to be projected on the embedding map. For example, a user might make up a new text that fits the explanation, and verify that it would fall in the cluster they’ve circled. Conversely, they might check that a text which would not fit the explanation would indeed fall outside the cluster. The point of these tests is to calibrate trust in the explanation. We call this workflow the **explain-and-test** framework. The rest of this section describes the TextCluster Explainer and the explain-and-test framework in detail.

3.1 Creating and Visualizing Text Embeddings

The explain-and-test framework can work with any text embedding method, but for the TextCluster Explainer prototype we choose the commonly used SentenceBERT package [23]. This system is free, achieves state-of-the-art sentence embedding performance on multiple tasks [24, 24], and has been used successfully in several other text visualization systems [8, 9]. The TextCluster Explainer reads a text file, breaks it into sentences, and then embeds these via SentenceBERT. For the purposes of our prototype, this provides a fairly general input format, but using other embedding tools it could easily be extended to work with larger pieces of text or collections of documents.

To visualize the embeddings, we use standard non-linear dimensionality reduction methods: t-SNE [29] and UMAP [19]. Users can choose either method to generate a projection view and start exploring the data via mouseover (Fig. 2).

3.2 Generating Text Explanations

Our main goal is to explain why a set of texts cluster in the visualization, i.e., what features set these data points apart from the rest. We compare explanations from a natural language interface and a baseline explainable method.

3.2.1 Baseline: Explanations via Keywords

Displaying the words or phrases that occur most frequently in the text can be a way to summarize it, however, keywords may be shared across multiple clusters and so may not be representative of a particular cluster. One way to mitigate this problem is to weigh the keywords according to how important they are for a particular cluster, as determined by learning a linear classifier based on word usage. For example, DeepClustering [10] obtained characteristic

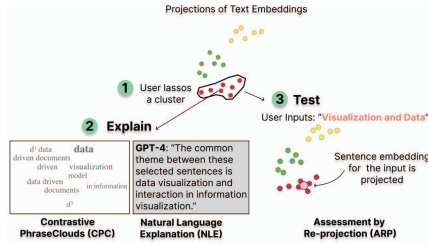


Figure 2: The basic workflow of our explain-and-test framework. A user can explore the pre-defined clusters of projections of text embeddings or lasso a region of interest to obtain explanations for the clustering in natural language using GPT-4 and as important phrases visualized as phraseclouds. New user input text can be embedded to test whether the explanations match a user’s mental model or discover topics of interest.

keywords by fitting a logistic regression model. However, these explanations contained unimportant words not related to the global cluster trend, most likely due to the imbalanced nature of the one-vs-rest classification task.

We build upon this and introduce *Contrastive PhraseClouds (CPC)*: visualizations of phrases found by fitting a support vector machine (SVM) model to the text inside a cluster and weighted by their relative importance (Supplementary material B).

The weights of the learned model lend themselves to a natural interpretation, with higher weights indicating higher importance for the cluster. We visualize top- n important phrases using word clouds, where the size of each word or phrase is scaled by its learned weight. This simple visualization conveys keywords in a familiar manner (Figure 1), but as with all word clouds puts a burden on the viewer to extract meaning [11].

3.2.2 Natural Language Explanations

AI systems based on large language models (LLMs) appear to excel at a wide variety of textual tasks [1]. A natural question is, can these systems be used to create concise, informative explanations for clusters of text embeddings? A key goal of the TextCluster Explainer is to explore such capabilities.

We integrate this capability into our prototype by allowing a user to request a natural language explanation of what the text in a selected cluster has in common. To create this explanation, we use the GPT-4 model from OpenAI [21]. We give the system the following prompt: “Please answer the following question without completing the sentences: What is the common theme between the following sentences:”. Prepending the prompt with the instruction to answer the question increases the tendency of the models to generate explanations rather than sentence completion. For advanced users, we offer the option of customizing this prompt. The sentences in a cluster are then appended to the prompt and passed to the GPT-4 version of the OpenAI Chat Completion API. The response is then presented to the user in an explanation panel on the side of the visualization. As a starting point for exploration, the framework can automatically cluster the projections and generate informative cluster labels (Supplementary material A). Example explanations of a number of lassosed clusters are shown in Figure 1 and Fig 4.

3.3 Testing Explanations: Assessment by Re-projection

Users don’t always trust AI-generated text—and with good reason. Neural network systems largely remain black boxes; GPT variants, in particular, are known to sometimes “hallucinate” responses [12, 15]. As a result, users will generally need to interact with the text to validate the generated explanations. Unfortunately, drilling down for details can be a cumbersome, lengthy process. Furthermore, it’s

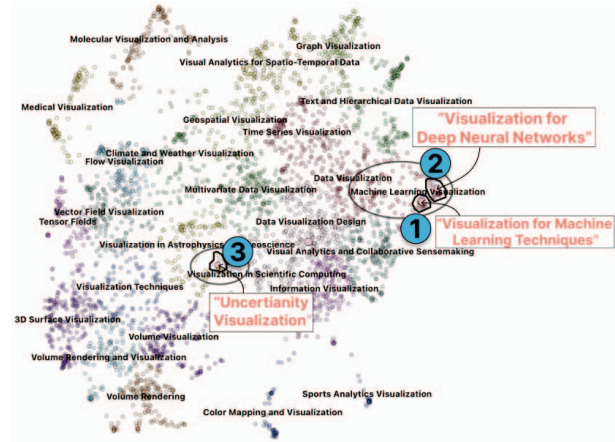


Figure 3: Natural Language Explanations can be used to generate summarizing labels for a cluster and capture a global trend in the text better than a set of keywords. This is extremely useful in generating cluster labels that summarize a group of text. The figure shows cluster labels overlaid on a UMAP projection of titles of IEEE VIS publications. Lassosed clusters 1-3 and *User inputs* further drive related topic discovery by facilitating the investigation of the region around a re-projected point.

often not enough just to examine elements of a cluster to see they fit the description; logically, one needs to check that elements outside the cluster do not fit the explanation.

To streamline the process of verifying the validity of cluster explanations, the TextCluster Explainer offers a feature that allows for rapid hypothesis testing. (This is the “test” part of “explain-and-test.”) A user can choose to enter a new, freeform piece of text, which is then “reprojected” onto the visualization. We call this “assessment by re-projection.”

When a user wants to validate a natural language explanation for a cluster, they can input custom text related to the explanation in the “Test new input” panel; this text is embedded and shown on the visualization. If the new projection lies inside the selected cluster, that adds support for the explanation. Alternatively, if the user enters text that does *not* fit the explanation, and it is reprojected outside the cluster, that too adds support for the explanation. As a bonus, this procedure can also be used to find an approximate region of interest for further exploration.

4 USAGE SCENARIOS

Here, we present two usage scenarios demonstrating the explanation generation capabilities of the TextCluster Explainer.

4.1 Investigating topics in academic paper titles

We use the TextCluster Explainer to extract research topics from paper titles in VisPub [13], a collection of IEEE Vis publications. Selecting the cluster labeled “Machine Learning Visualization,” the Contrastive PhraseCloud produces the phrases: {*classifiers, deep, neural networks, explainable, decision trees*}. The natural language explanation is: *The common theme is visualization, analysis, and interpretation of machine learning models, particularly deep learning and neural networks*. While this is a well-written explanation, a user might not know whether to trust it—or perhaps want clarification, such as whether the topic of explainable AI would fall in this cluster.

Taking advantage of the explain-and-test interface, the user tests the phrase *Visualization for Explainable AI* and finds it within the selection (subcluster (1) in Fig. 3). The explanation for this subcluster is: *The common theme is visual analytics, interactive methods,*

and explainable techniques in machine learning and artificial intelligence, particularly focusing on classifiers, decision-making, and understanding models. Thus the test input guides the user when the natural language explanation is verbose but not comprehensive, demonstrating its utility.

This drives further topic exploration as the user notices the other high-density (subcluster (2) in Fig 3), with explanation: *The common theme is visualizing, analyzing, and understanding deep neural networks and their components using various tools and techniques.* The user can surmise that subcluster (1) is related to works visualizing and explaining machine learning techniques, while subcluster (2) is related to works on visualization for deep learning models. However, one might be wary of outputs generated from current LLMs and might want to confirm this hypothesis. By re-projecting the user input *Visualization for Deep Neural Networks*, one finds the projected points corresponding to these texts lie exactly on subcluster (1).

The user may also want to explore topics related to visualizing uncertainty. They start by entering *Uncertainty Visualization* as an example input and discover it lies in cluster (3). (This illustrates how reprojection can work as a search tool.) Lassoing the cluster, the user finds related phrases from the PhraseClouds: *uncertainty, geospatial modeling, modeling system, tangible geospatial, scientific visualization, bounded uncertainty* and the natural language explanation: *The common theme is uncertainty visualization, scientific visualization, or data visualization in various domains such as archaeology, anthropology, and geodemographics.* This points to works visualizing uncertainty in specific areas and the general field of scientific visualization. The explanation again is informative, though an expert reader may note it juxtaposes scientific visualization with smaller subfields of visualization. TextCluster Explorer encourages a user to interact with text data and discover new insights.

4.2 Literary Text Exploration

In our second example, we show how the TextCluster Explorer can guide exploration of a literary text. Here the input is “Frankenstein” by Mary Shelley [26]; the visualization shows all sentences in the book (Figure 4). Without further interaction, the labels on automatically generated clusters appear to distill the content of the work into different overarching thematic and narrative sections.

For example, sentences in the cluster related to *Natural Philosophy and Science* describe Victor Frankenstein’s interactions with two professors of Natural Science and Philosophy and their influence on him. The key phrases obtained from the phraseclouds: *science, laboratory, discovery, krempe, waldman, laboratory, natural philosophy, professors* (4) support this. In contrast, in the *Dream-like state* cluster points to the parts of the work surrounding the dream-like state that Frankenstein found himself in while creating his monster and after he saw his creation come to life [6]. Themes of isolation are prominent in the text. On investigating the region surrounding the projected user input: *Isolation and Belonging*, this phrase lies close to the clusters labeled *Search for acceptance and companionship*, *Despair and Misery* and *Remorse and Suffering*. A user can explore these further as necessary.

Interaction can reveal further insights. Fig. 4 shows examples of selected clusters and their natural language explanations. The explanations seem to relate to key themes in the book as well as central characters. Indeed, working with selections uncovers an interesting pattern: tight clusters general refer to specific characters or groups of people. The explanations are also surprisingly nuanced, as for a cluster described as relating to “mentors or teachers.” Such as description goes beyond what a list of keywords can offer.

5 CONCLUSION AND FUTURE WORK

We have proposed a framework, **explain-and-test**, for working with visualizations of text embeddings. In this framework, AI-generated “explanations” are paired with a way of verifying them through

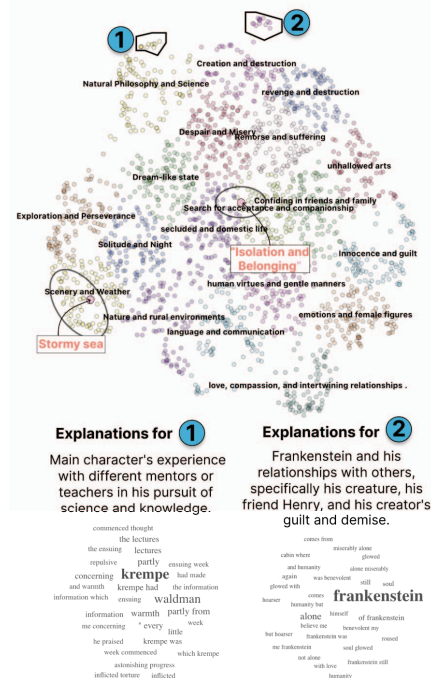


Figure 4: Natural Language Explanation labels can also capture a global semantic trend of the text in a cluster. This is important for exploring literary works. The figure shows cluster labels overlaid on a UMAP projection of sentences from “Frankenstein” by Mary Shelley [26]. User inputs allow exploration of the themes and characters in this work.

user-supplied “test” examples. The goal is to exploit the powerful capabilities of machine learning while addressing the basic fact of life with AI: explanations generated via machine learning are not always reliable.

To demonstrate this framework, we built an application, the TextCluster Explorer. The tool includes two types of cluster explanations: one based on a “white box” linear classifier, and one based on a “black box” state-of-the-art language model system, GPT-4. Using this system, we presented usage scenarios that illustrate the explain-and-test framework. These preliminary results also indicate that GPT-4 provides nuanced and comprehensible cluster descriptions, with the potential to be an important ingredient in future visualization tools. We illustrate two applications utilizing short text; however, the proposed framework’s modular design can accommodate extensive text lengths. Using an LLM with a larger context window ([27, 28]), users can analyze longer texts, extending from comprehensive academic papers to entire documents or books.

Several directions for future research look promising. It would certainly be informative to do user studies to verify the efficacy of the explain-and-test framework, especially in the context of the TextCluster Explorer. More generally, given the apparent power of GPT-4 to produce fluent explanations, there is a great deal of space to explore ways to automate visual text analytics, from labeling regions of a visualization, to discovering outliers, and finding global patterns.

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